

T-MATRIX PIPELINE · METHOD, VALIDATION, AND A DIAGNOSED LIMIT

Composing Metasurfaces from Measured Meta-Atoms

Extract each meta-atom's transition matrix from one full-wave simulation, then predict whole metasurfaces by linear algebra — where that works, where it breaks, and why the break is not fixable inside the aggregation algorithm.

D:\Claude\T matrix · ten direct-CST benchmarks · cross-verified against treams (KIT) to 10^{-12} · August 2026

One full-wave solve per atom — then metasurfaces become linear algebra

1 Extract

once per meta-atom

- One CST solve of the isolated atom (PML, plane-wave illumination set)
- Near fields sampled on a sphere, projected onto vector spherical waves (VSWFs)
- Least-squares fit $F = T \cdot A \rightarrow$ the atom's transition matrix T (30×30 at $l_{\max} 3$)

2 Aggregate

per arrangement

- Place measured atoms on any periodic cell — one species or several
- Pair-resolved Ewald lattice sums couple every atom to every other, exactly
- Solve the block Foldy–Lax system for the self-consistent excitations

3 Predict

milliseconds each

- Project onto Floquet orders $\rightarrow$ full complex S-parameters, all diffraction channels
- No full-wave simulation of the composed array is ever run
- 64 min of CST per supercell benchmark vs milliseconds per composed design

The promise: simulate each atom once, then compose arbitrarily many metasurfaces from the measured parts — the repeated cell may hold one meta-atom or several different ones.

The algebra: block-Bloch Foldy–Lax with Ewald lattice sums

Atom response (VSWF basis)

$$p = T a$$

Pair-resolved Bloch lattice sum (Ewald)

$$W_{st}(k//) = \Sigma' R A(\rho_t - \rho_s + R) \cdot \exp(i k// \cdot R)$$

Self-consistent block solve, M atoms per cell

$$(I - W T) a = a_{inc}, \quad f = T a$$

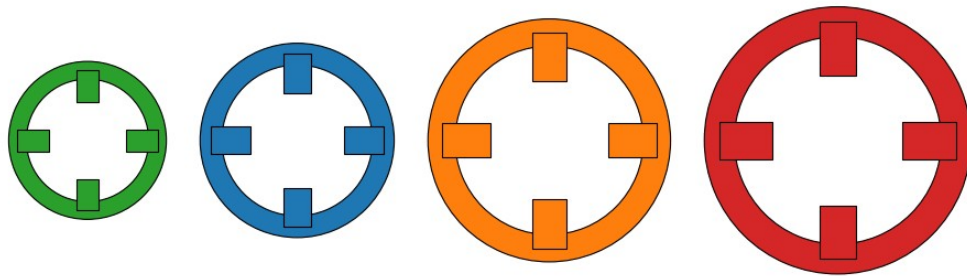
Floquet output map, order G on either side

$$E_G \propto (1/k_{z,G}) \Sigma_s \exp(-i k_{z,G} \cdot \rho_s) F_s(\hat{k}_G)$$

- Basis: vector spherical waves to $l_{\max} 3 \rightarrow 30$ modes per atom; a four-atom cell is a dense 120×120 solve.
- Lattice sums by Ewald summation (via treams); an η -stability bracket refuses rather than guesses when the split disagrees.
- Conventions measured identical to treams: VSWFs to 7×10^{-16} , plane-wave expansion 6×10^{-16} , translation operator 5×10^{-15} .
- Every propagating diffraction order is returned, power-normalized — supercells diffract inside the band.

Four measured gold meta-atoms, ten blind CST benchmarks

the four meta-atoms, to scale



C
scale 3.25
 $r = 2.34 \mu\text{m}$

A
scale 4.00
 $r = 2.88 \mu\text{m}$

B
scale 5.00
 $r = 3.60 \mu\text{m}$

D
scale 5.50
 $r = 3.96 \mu\text{m}$

The four spoke-and-wheel resonators, drawn to scale. All are $0.2 \mu\text{m}$ -thick gold, extracted in isolation over 10–34 THz (25 frequencies, stored at $I_{\text{max}} 5$).

atom	scale	ring outer radius	isolated resonance / scale
C	3.25	$2.338 \mu\text{m}$	3.766
A	4.00	$2.877 \mu\text{m}$	3.770
B	5.00	$3.596 \mu\text{m}$	3.786
D	5.50	$3.956 \mu\text{m}$	3.782

- Layout: $16 \times 16 \mu\text{m}$ repeated cell, atoms on the $8 \mu\text{m}$ square sub-lattice — four single-atom lattices + six mixed supercells.
- Every case has an independent direct-CST periodic reference: ten blind benchmarks in all.
- Input imperfections are measured, not assumed: the extracted T's violate passivity by 2–5 % and reciprocity by up to ~18 % — the inputs, not the aggregation, set the error floor of the good cases.

The aggregation itself is exact to round-off

algebraic identity / independent implementation	residual
M = 1 cell reduces to the one-atom code (coupling, f, S11/S21)	bit-identical
2×2 cell of four identical atoms $\equiv$ the primitive lattice	$\leq 8 \times 10^{-16}$
basis-atom relabelling / whole-cell lattice shift	$\leq 7 \times 10^{-16} / 0$
checkerboard selection rule: power in odd (n_1+n_2) orders	$\leq 6 \times 10^{-33}$
all $T = 0 \rightarrow S = S_{\text{background}}$ exactly	0
independent treams implementation of the whole chain, every cell, complex S	$\leq 1 \times 10^{-12}$

What this rules out

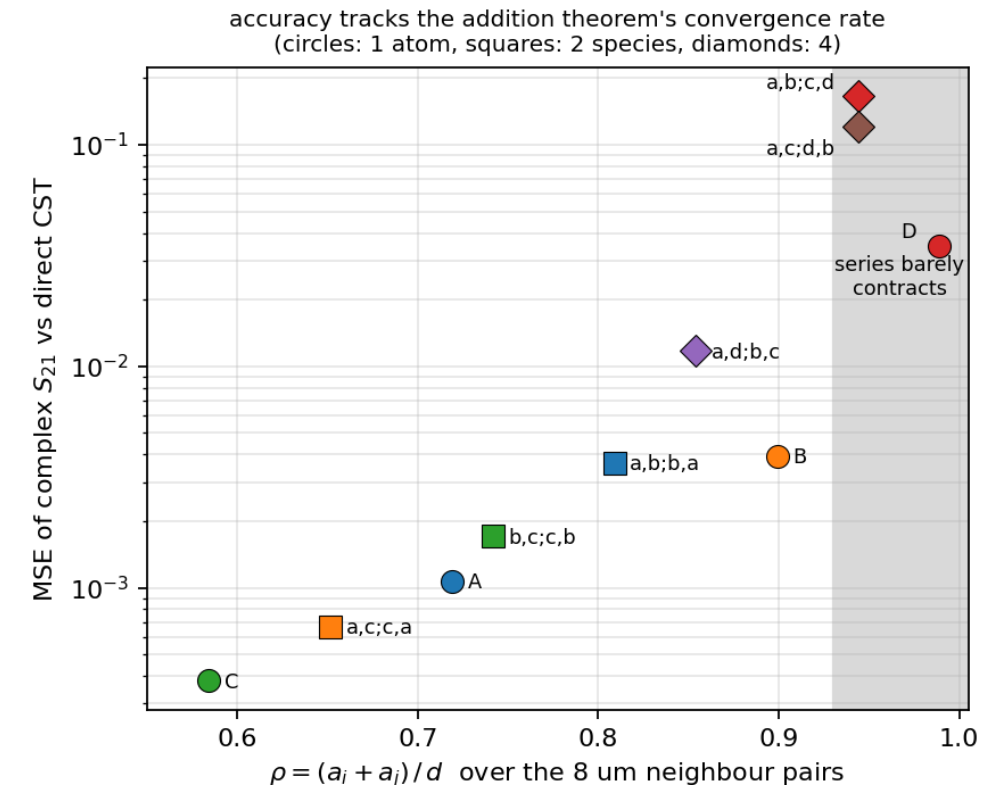
- coding errors in the block solve
- lattice-sum convergence
- basis conventions and normalization
- the Floquet output map

Consequence: whatever error remains against full-wave CST lives in what a truncated single-centre T-matrix can represent — not in the code that manipulates it.

Accuracy is ordered by one geometric number, $\rho = (a_i + a_j)/d$

case	ρ (worst pair)	MSE of complex S_{21}	mean $ \Delta S_{21} $	verdict
C alone	0.584	0.00038	0.017	good
a,c;c,a	0.652	0.00067	0.019	good
A alone	0.719	0.00107	0.030	good
b,c;c,b	0.742	0.00171	0.036	good
a,b;b,a	0.809	0.00367	0.046	good
a,d;b,c	0.854	0.0118	0.080	degrading
B alone	0.899	0.0039	0.054	degrading
a,c;d,b	0.944	0.1207	0.244	broken
a,b;c,d	0.944	0.1654	0.308	broken
D alone	0.989	0.0351	0.149	broken

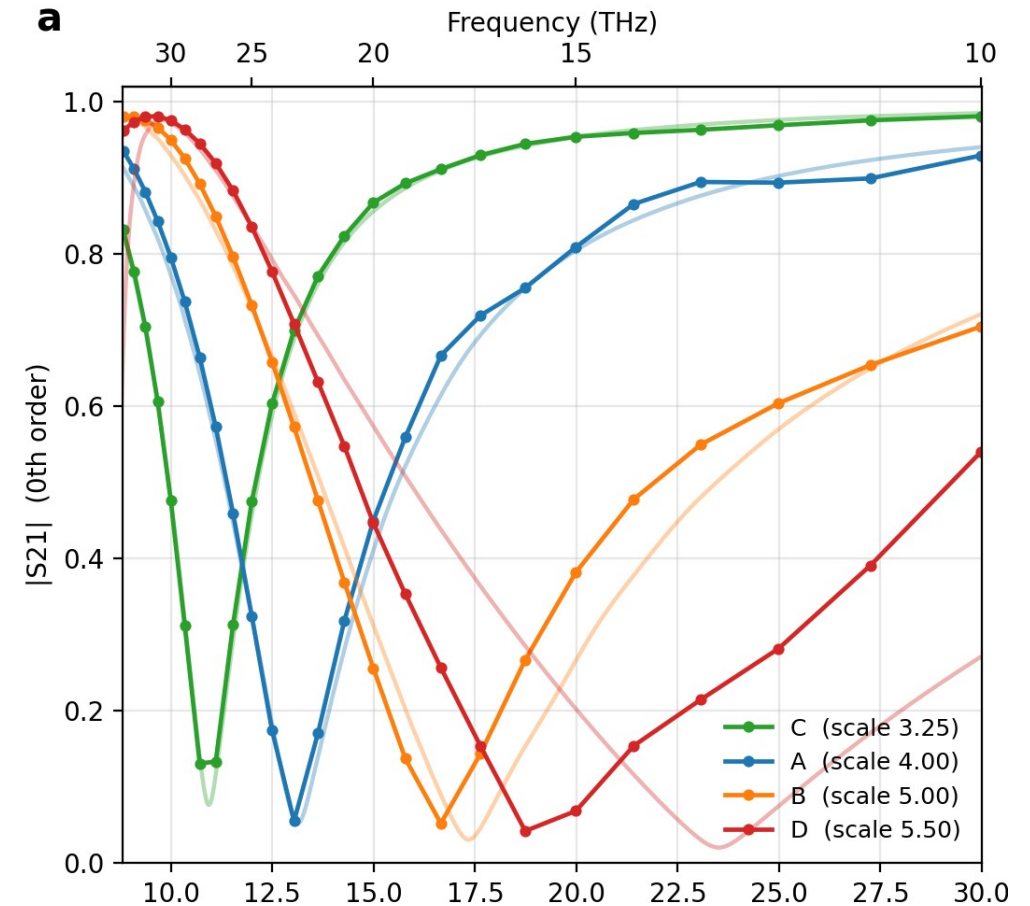
MSE = mean $|S_{21_pred} - S_{21_CST}|^2$ over the band, on the complex amplitude, 0th order. Seven of ten land at mean $|\Delta S_{21}|$ 0.017–0.080 — the level set by the input T-matrices.



Same data: MSE vs ρ . Above $\rho \approx 0.94$ (shaded) the multipole series barely contracts.

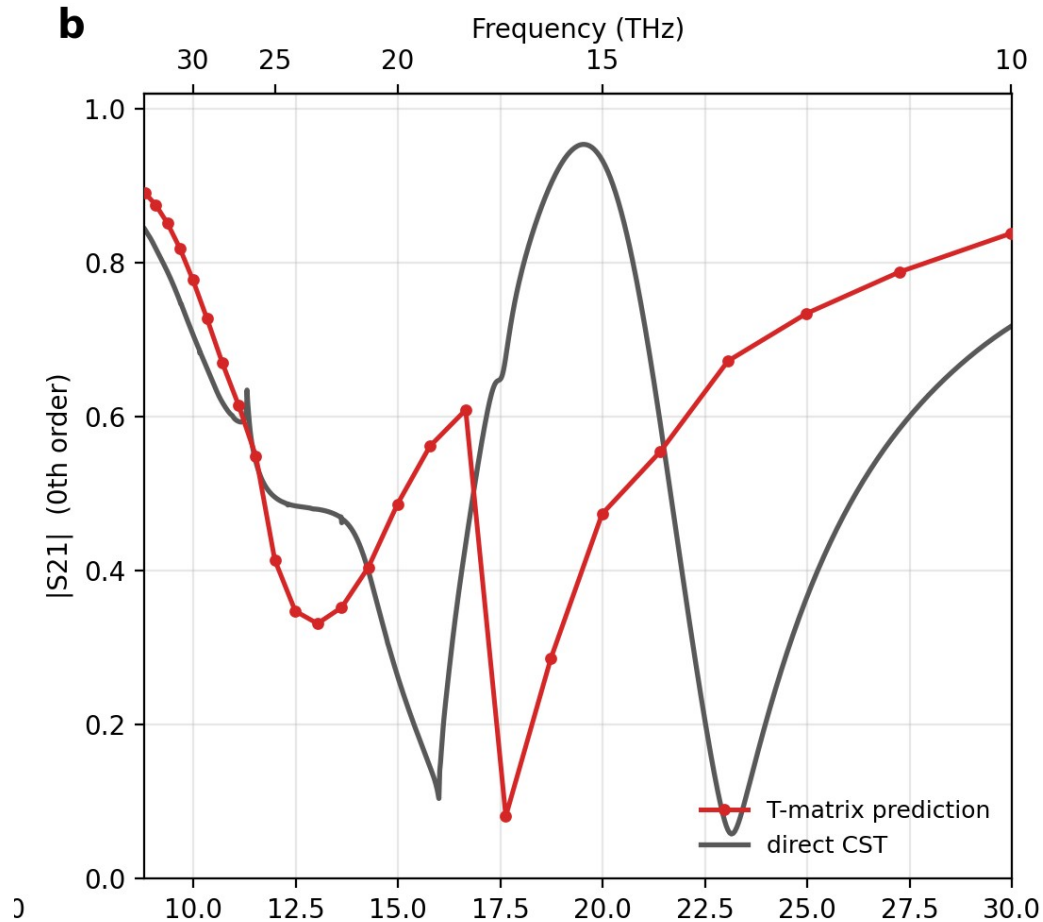
Where ρ is moderate, composition is essentially free

- Seven of ten benchmarks: mean $|\Delta S_{21}|$ 0.017–0.080 — within the input T-matrices' own passivity / reciprocity error.
- The supercell step adds nothing: a,b;b,a lands at 0.046, between its two pure lattices (0.030 and 0.054).
- A dark lattice resonance at 16.00 μm — invisible to any single-atom model — was predicted and then confirmed by direct CST: dark-channel power 1.5×10^{-5} against a 0.987 carrier.
- Its transmission window: predicted 15.57 μm / $|S_{21}|$ 0.68, measured 15.35 μm / 0.72.
- Exact selection rule (odd diffraction orders extinguished) confirmed in every cell.



One atom per 8 μm cell: prediction (markers) vs direct CST (pale). C, A, B track closely — D, the largest, already fails. That failure is the subject of the rest of this deck.

Same atoms, same lattice — 14× worse when B and D become neighbours



a,b;c,d: predicted $|S_{21}|$ (red) vs direct CST (grey). The deepest dip is misplaced by $-5.51 \mu\text{m}$.

MSE 0.1654

vs 0.0118 for a,d;b,c — identical atoms, identical lattice, different adjacency

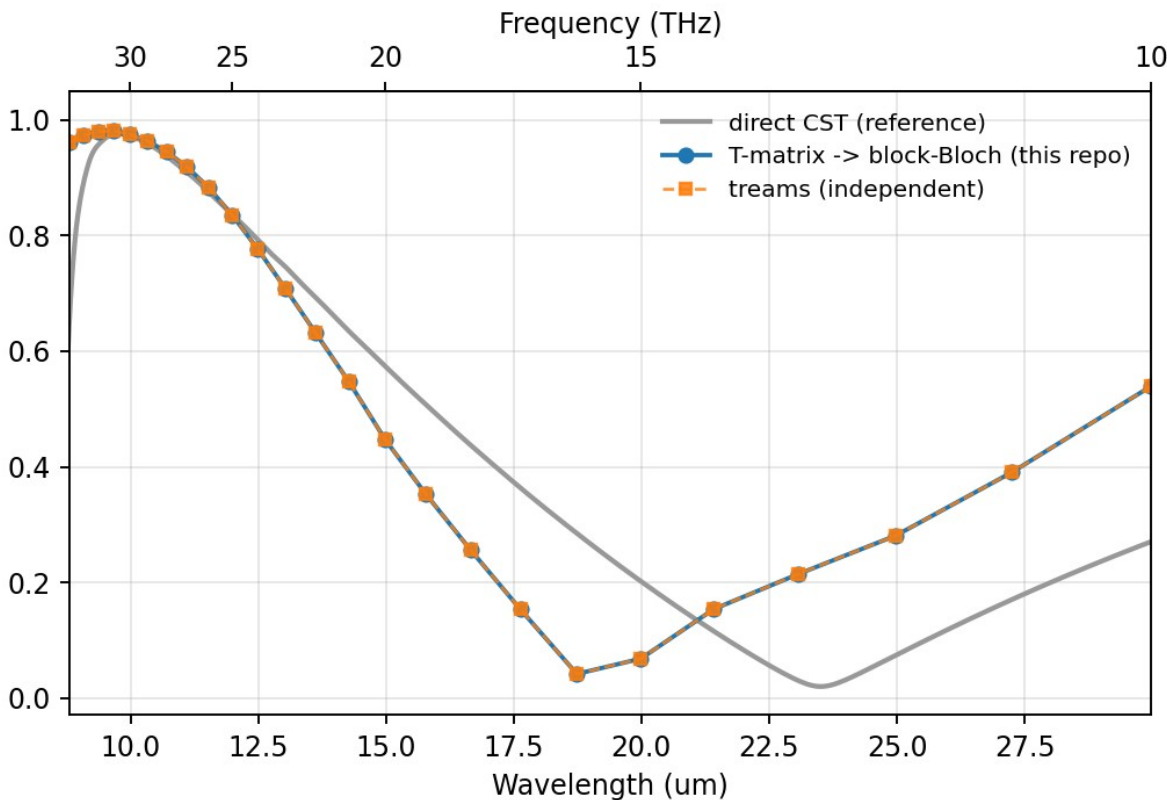
$-5.51 \mu\text{m}$ / $+4.92 \mu\text{m}$

dip misplacement in a,b;c,d and its p-twin a,c;d,b — opposite signs, so no calibratable bias

0.409 vs 0.388

total diffracted power of a,b;c,d, predicted vs measured (5.4 %); the p-twin a,c;d,b keeps 1.0 % (0.407 vs 0.403) — the specular channel breaks first

Atom D alone: the resonance lands 19 % away — and red-shifted



D alone on its 8 μm lattice: predicted dip 19.1 μm (repo = treams, indistinguishable) vs 23.5 μm measured; absorption goes negative.

- A, B, C blue-shift from isolated to array resonance by a consistent 1.3–2.0 μm. D red-shifts by +2.74 μm — opposite sign.
- Across the parametric sweep the anomaly tracks the closing gap, not the atom:

scale	gap to neighbour	CST array dip	dip / scale
4.00	2.246 μm	13.13 μm	3.283
4.50	1.526 μm	14.90 μm	3.311
5.00	0.807 μm	17.34 μm	3.468
5.50 (= D)	0.088 μm	23.51 μm	4.275

88 nm apart at scale 5.5: the classic near-touching capacitive red-shift — a hybridized gap mode of the pair. Real physics, present in full-wave, absent from the prediction.

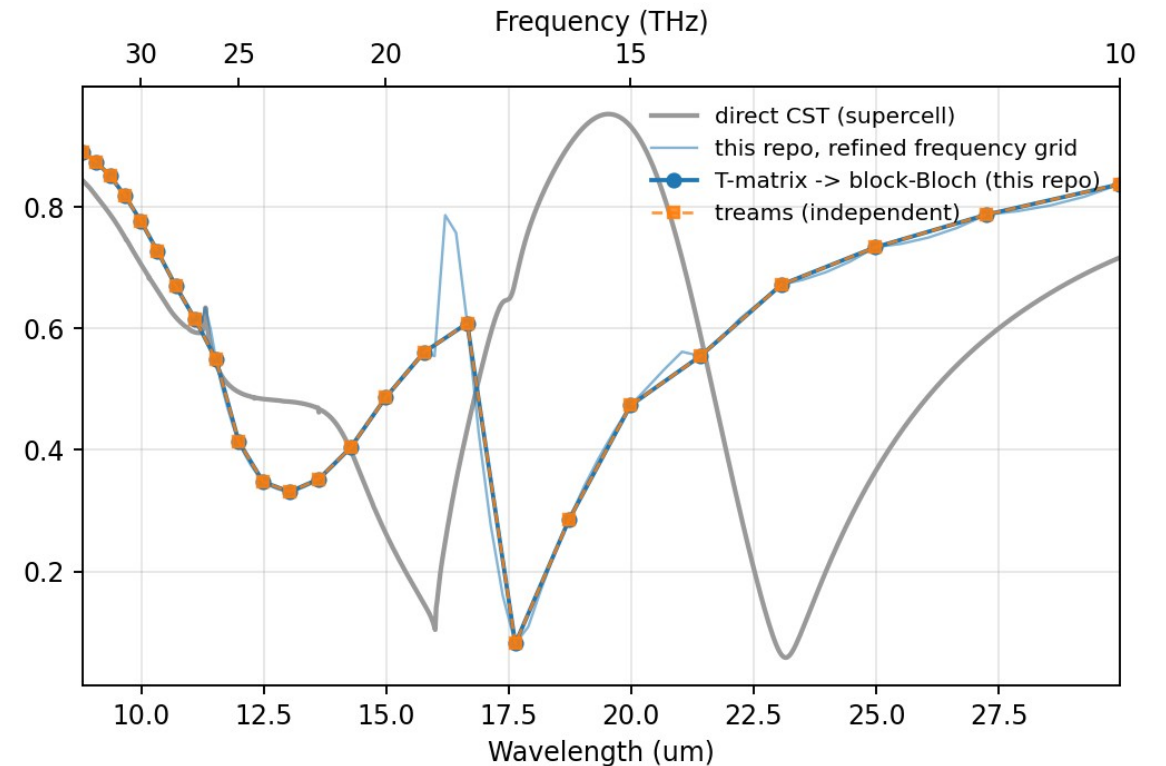
The T-matrix is innocent — and so is the implementation

The input T-matrix is consistent

- Isolated resonance / scale: 3.766, 3.770, 3.786, 3.782 for C, A, B, D — perfectly linear; D is no outlier.
- D is the most dipolar of the four (96.5 % $l = 1$), so truncating its own multipoles is not the issue.

The code is consistent — twice over

- treams — an independent implementation — reproduces D's wrong answer to 1.2×10^{-15} .
- It evaluates the same addition theorem at the same $l_{\max}$, so it inherits the same limit.



a,b,c,d: this repo (blue) and treams (orange) are indistinguishable — and both sit far from CST (grey).

Cross-code agreement validates the implementation. Only full-wave validates the physics.

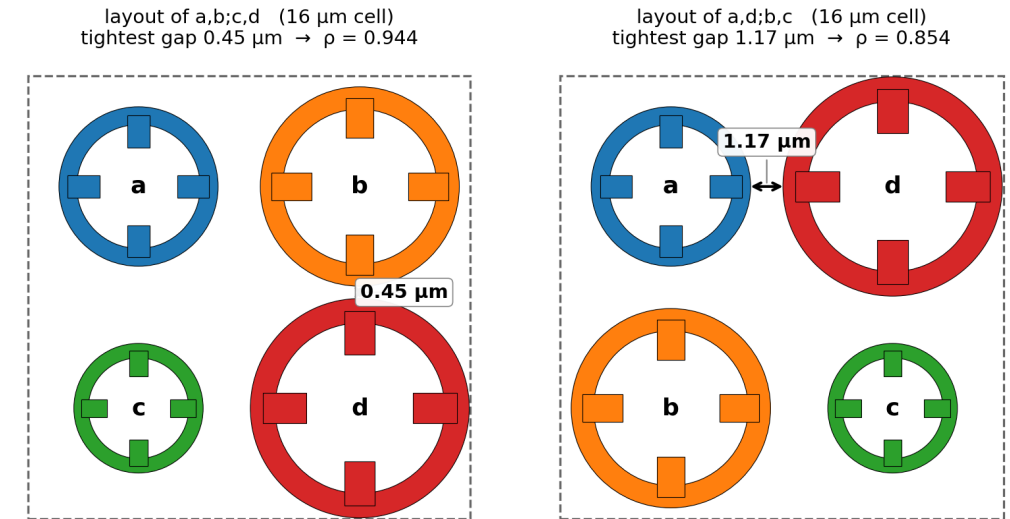
Truncated addition theorem: error contracts by only ρ per order

- A T-matrix describes its atom only outside the circumscribing sphere (radius a).
- Re-expanding neighbour j 's scattered field about atom i converges like a geometric series in $\rho = (a_i + a_j)/d$ — truncation at $l_{\max}$ leaves an error $\sim \rho^{l_{\max}}$.

ρ	case	$l_{\max}$ needed for ~1 %
0.652	a,c;c,a	11
0.809	a,b;b,a	22
0.944	a,b;c,d (B–D pair)	80
0.989	D alone (D–D pair)	416

Used: $l_{\max}$ 3.

Silent by construction: the validity condition $a_i + a_j < d$ is satisfied in every case (D–D: 7.912 vs 8.000 μm). Marginally satisfied is worse than violated — nothing warns.



The knob that matters: a,b;c,d puts B and D adjacent (gap 0.45 μm , $\rho = 0.944$); a,d;b,c repairs the same four atoms (tightest gap 1.17 μm , $\rho = 0.854$) and is 14× more accurate.

Raising $l_{\max}$ makes it worse: two errors move in opposite directions

↓ truncation error

- Falls like $\rho^{l_{\max}}$ — at $\rho = 0.944$ each extra order buys a factor 0.944: $l_{\max} \approx 80$ for 1 %.
- At $l_{\max}$ 3–5 the neglected remainder of the B–D coupling is of order 1.

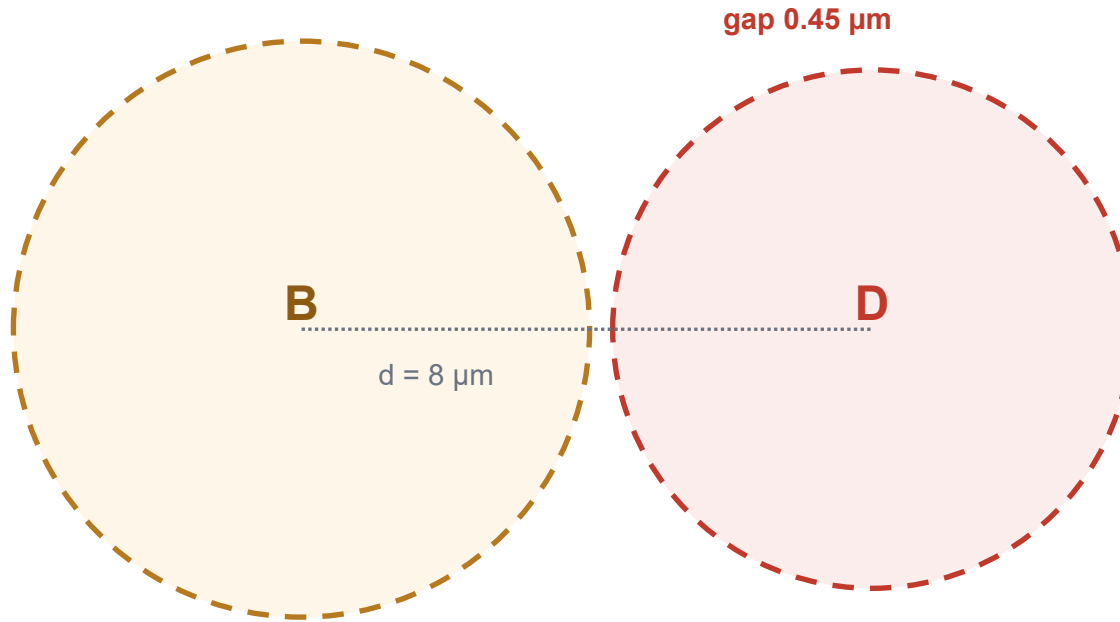
↑ noise amplification

- The $l = 4, 5$ rows of the measured T sit at the extraction noise floor.
- The near-field lattice sum amplifies exactly those rows: $h_l(kp) \sim (2l-1)!! / (kp)^{l+1}$ — $|W|$ at $l = 3$ reaches 1025 at $\lambda \approx 30 \mu\text{m}$ vs 1.7 at $8.8 \mu\text{m}$: 227× the dipole coupling there.

Both measured, not predicted:

- Atom D's reconstruction is worse at $l_{\max} 5$ than at $l_{\max} 3$ — the noise term already dominates before truncation is fixed.
- The obvious internal check fails too: the $l_{\max} 3 \rightarrow 4 \rightarrow 5$ spread is largest for the most accurate cell $(a,d;b,c)$, because the spread measures noise amplification, not geometry. There is no usable convergence warning.

The limit is representational, not algorithmic



each T-matrix is valid only outside its dashed sphere — the gap region belongs to neither expansion

- A single-centre, $l_{\max}$ -3 T-matrix has 30 modes. The field in a $0.45\text{ }\mu\text{m}$ gap varies on the scale of the gap — that information does not exist in those 30 numbers.
- ρ is fixed by geometry: circumscribing radii and centre distance. No solver setting, no lattice-sum method, no basis convention changes it.
- The hybridized gap mode (the measured red-shift) is reachable by the formalism only at $l_{\max} \approx 80\text{--}400$ — far beyond what measured T-matrices support.

No modification of the aggregation alone can supply the missing physics

Proof in one line — the Neumann sandwich

$$f = T a + T C T a + T C T C T a + \dots$$

- Every lattice operator C is sandwiched between two copies of the stored T . T is zero outside $l \leq 3$, so any higher-order content added to C is annihilated on contact — computing C exactly, at any $l_{\max}$, changes nothing.

The other obvious fix also dies — on geometry

Merging the near-touching pair into one extracted super-atom makes a sphere that swallows the next neighbour:

merge	merged radius	worst ρ vs rest
B + D	7.956 μm	1.211 — invalid
A + B	7.596 μm	1.292 — invalid
C + D	7.956 μm	1.292 — invalid

($\rho \geq 1$ violates the validity condition outright — merging only works in sparse cells)

Therefore

The missing gap-scale information must enter upstream of the solve: the representation of each atom — i.e. the extraction — has to change, not the aggregation that consumes it.

Multi-centre representation: make ρ an algorithmic choice again

- Replace one $4\text{ }\mu\text{m}$ expansion centre by ~ 30 sub-centres of radius $r_s \approx 0.5\text{ }\mu\text{m}$ tiling the ring. Since $k \cdot r_s \approx 0.24$, each needs only dipoles ($l_{\text{max}} 1$): ~ 180 modes per atom, still a trivial dense solve.
- The pair convergence ratio becomes $\rho_{\text{sub}} = 2r_s / (g + 2r_s)$ — set by a chosen radius, no longer by the fixed geometry.

pair	gap g	ρ today	ρ_{sub} at $r_s = 0.5\text{ }\mu\text{m}$
B–D	$0.448\text{ }\mu\text{m}$	0.944	0.691
A–D	$1.167\text{ }\mu\text{m}$	0.854	0.461
D–D	$0.088\text{ }\mu\text{m}$	0.989	0.919 (graded radii near the gap)

B–D at $\rho_{\text{sub}} = 0.691$ would sit among the best-performing cells of the study ($\rho \approx 0.65\text{--}0.75$).

~80 % already built

- Block lattice sums take arbitrary per-cell positions (treams shift list)
- Multi-centre cluster operators exist (cluster_T)
- Change needed: dense per-atom blocks instead of diagonal T's — a data-model edit

The real cost: extraction

- ≥ 180 independent excitations per atom
- Plane waves are too smooth — near-field (dipole) sources must join the illumination set
- A new CST protocol; the existing conditioning machinery is the right design tool

What to do while the representation is rebuilt

Gate on ρ

Warn above 0.85, refuse above 0.95 (with --force). The η -split already refuses rather than guesses — extend the policy. Blocked only on storing the circumscribing radius in tmat.h5.

Design around it

Never place the two largest atoms adjacent: a,d;b,c vs a,b;c,d is 14× better for free — same atoms, same lattice.

Denoise, then $I_{\max}$

Symmetry-average the noisy $I = 4, 5$ rows before any $I_{\max}$ raise. Helps the $\rho \approx 0.85$ band; cannot reach 0.944.

1

Validated: composing metasurfaces from measured atoms is essentially free below $\rho \approx 0.85$ — accuracy is set by the inputs, not the method.

2

The failures (a,b;c,d, a,c;d,b, atom D) are one diagnosed mechanism — slow convergence of the truncated addition theorem across sub- μm gaps — not noise, and not a bug: treams reproduces them to 10^{-15} .

3

The fix is representational, not algorithmic: the aggregation provably cannot supply the missing gap physics; a multi-centre extraction can.